

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 10

INSOLATION

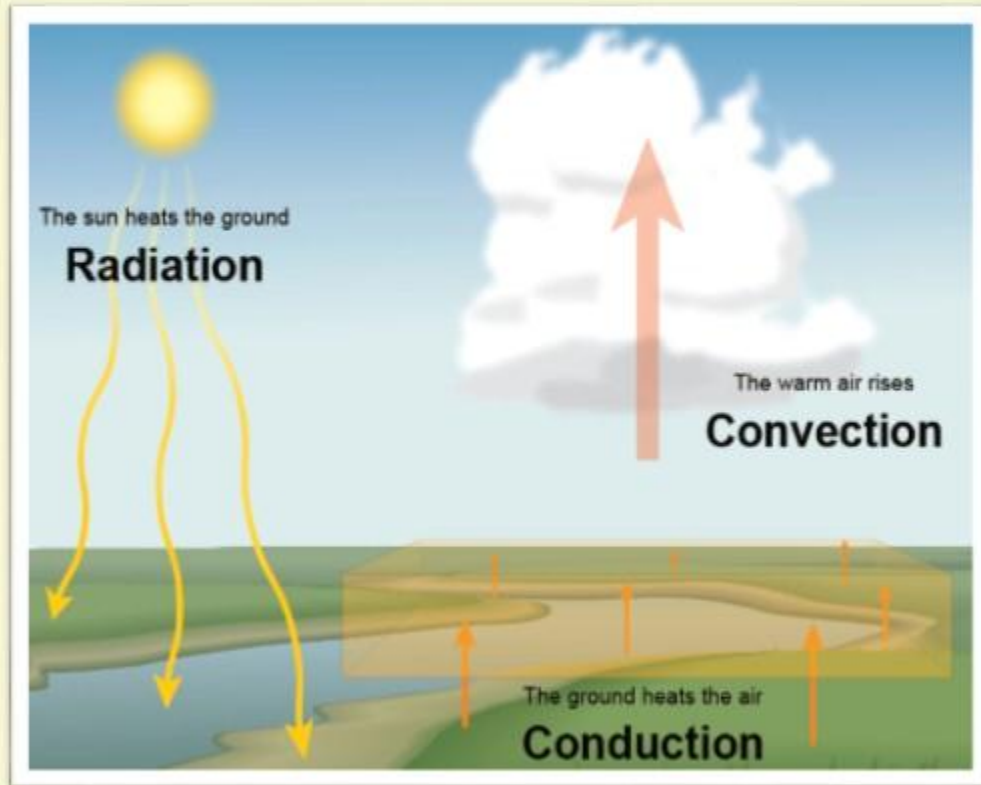
- HEAT BUDGET
- HEATING & COOLING OF ATMOSPHERE



HEATING AND COOLING OF THE ATMOSPHERE



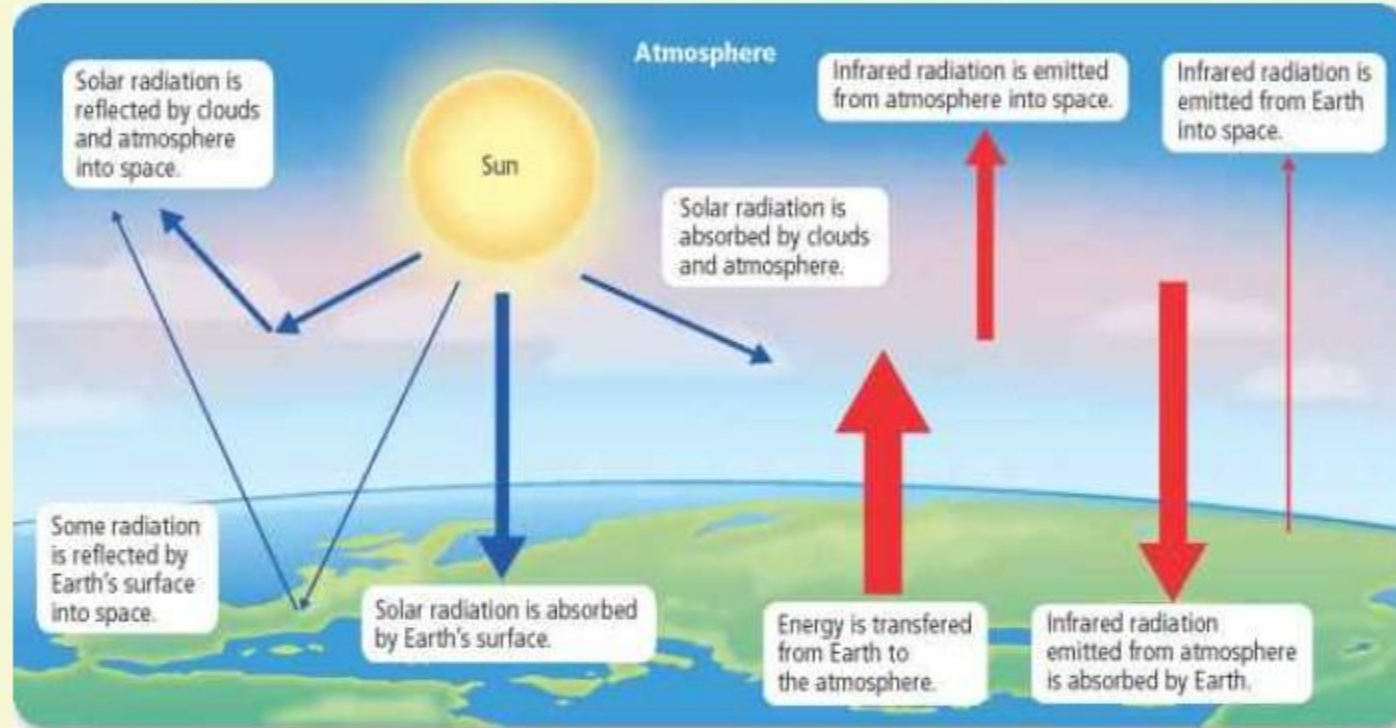
CONDUCTION



- The Earth after getting heated by insolation transmits the heat to the lower layers of atmosphere which is in contact with it. This process is called conduction.
- Conduction takes place when two bodies of unequal temperature are in contact with one another, there is a flow of energy from the warmer to cooler body.
- The transfer of heat continues until both the bodies attain the same temperature or the contact is broken.
- Conduction is important in heating the lower layers of the atmosphere

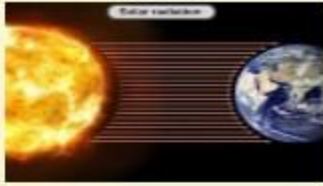


HEATING AND COOLING OF THE ATMOSPHERE

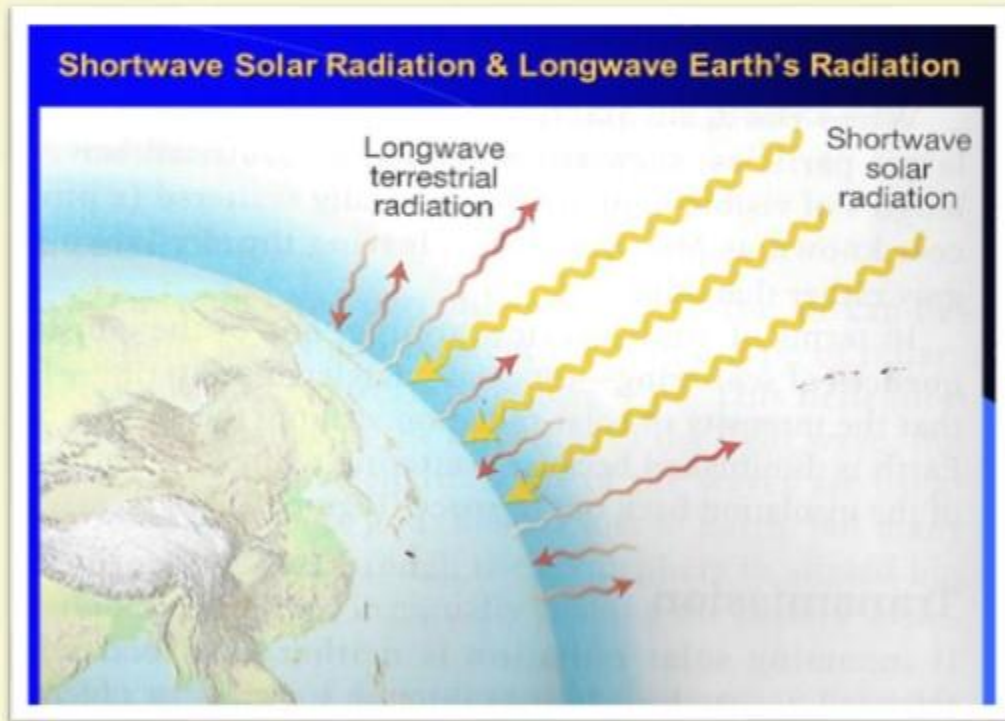


CONVECTION

- The air in contact with the earth rises vertically on heating in the form of currents and further transmits the heat of the atmosphere.
- This process of **vertical heating of the atmosphere is known as convection**.
- The convective transfer of energy is confined only to the troposphere.



HEATING AND COOLING OF THE ATMOSPHERE



RADIATION

- The insolation received by the earth heats up its surface.
- The earth after being heated becomes a **radiating body** in itself and it radiates energy back to the atmosphere in long wave form.
- This energy heats up the atmosphere from below. This process is known as **terrestrial radiation**.

Why is the outgoing radiation in the long-wave form while the incoming rays are short-wave radiation?



15 million °C



15 °C



High Energy = High Frequency Waves

High Frequency = Low Wavelength

Thus, Sun emits
SHORTWAVE RADIATION



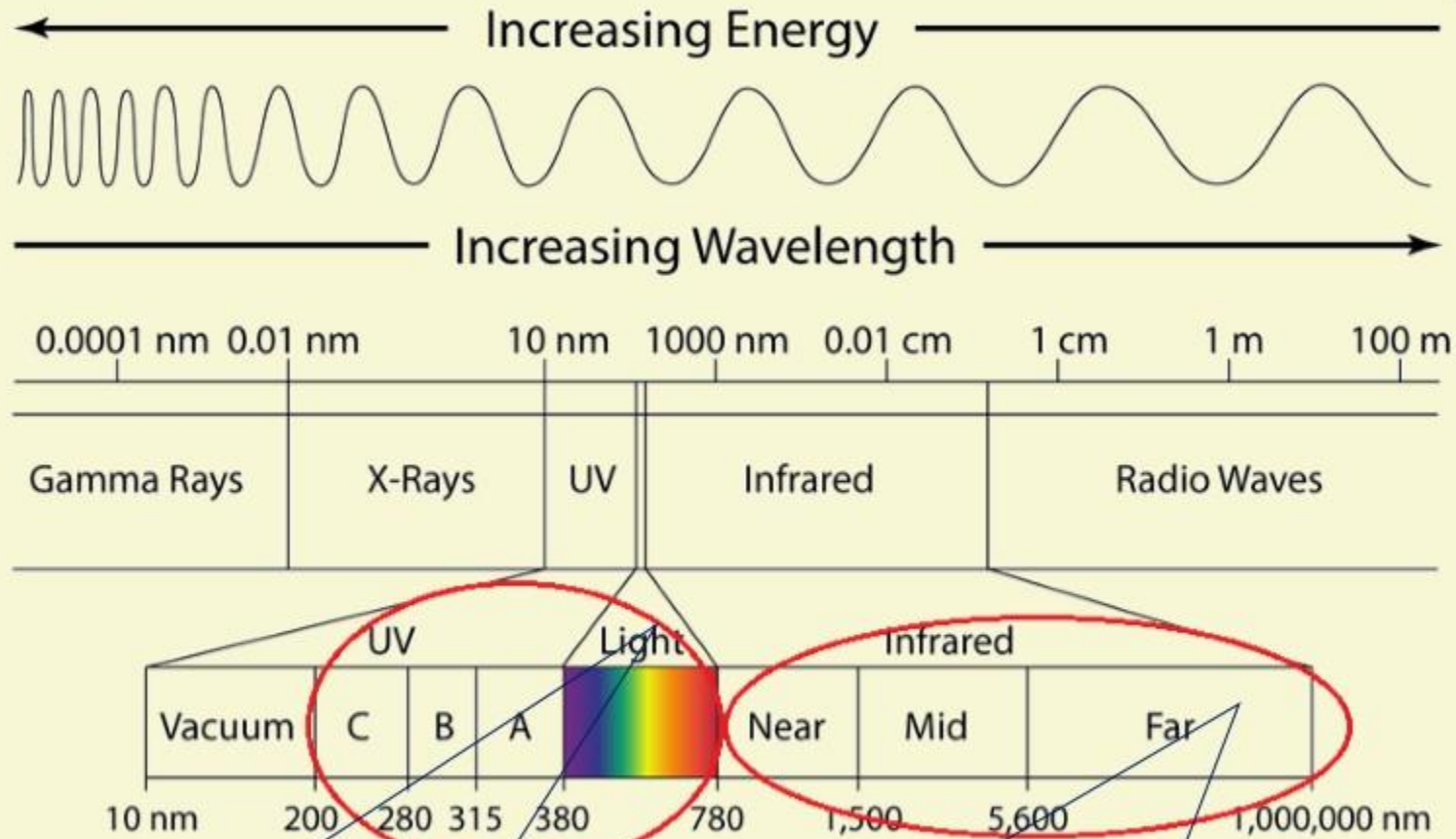
Low Energy = Low Frequency Waves

Low Frequency = High Wavelength

While, Earth emits
LONGWAVE RADIATION



Why is the outgoing radiation in the long-wave form while the incoming rays are short-wave radiation?



The Sun's incoming shortwave radiation is mostly in this zone of UV light and visible light

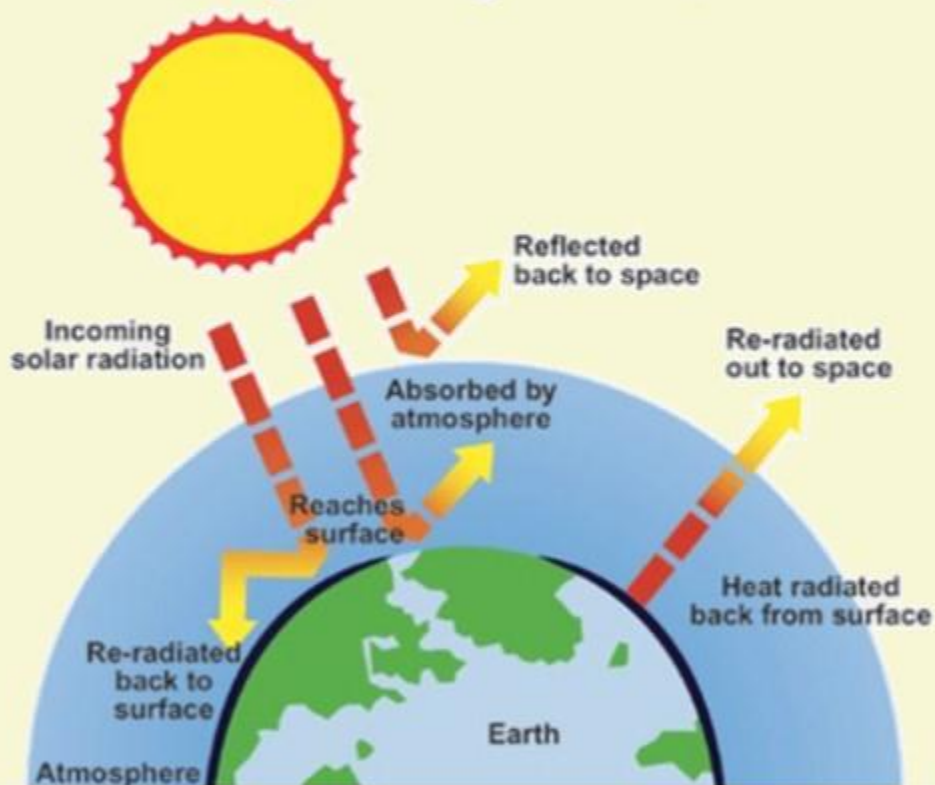
The Earth emits outgoing longwave radiation mostly in this zone of infrared light (composed mostly of heat)



HEATING AND COOLING OF THE ATMOSPHERE



Global warming and the greenhouse effect



COUNTER-RADIATION

- Some part of the ground radiation after being absorbed by the atmosphere is **radiated back to the earth's surface**.
- This process of radiation of terrestrial heat energy from the atmosphere back to the earth's surface is called **counter-radiation** which is effected mainly by **water vapour and carbon dioxide**.

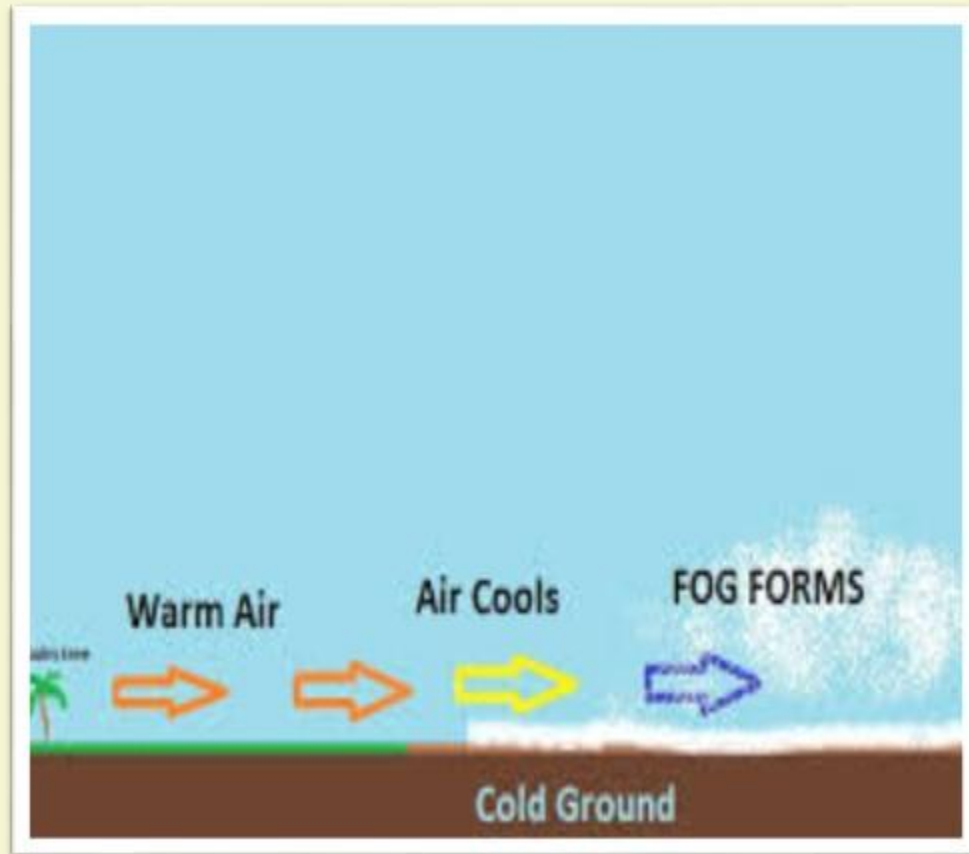


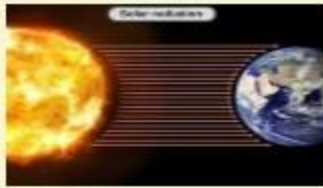
HEATING AND COOLING OF THE ATMOSPHERE



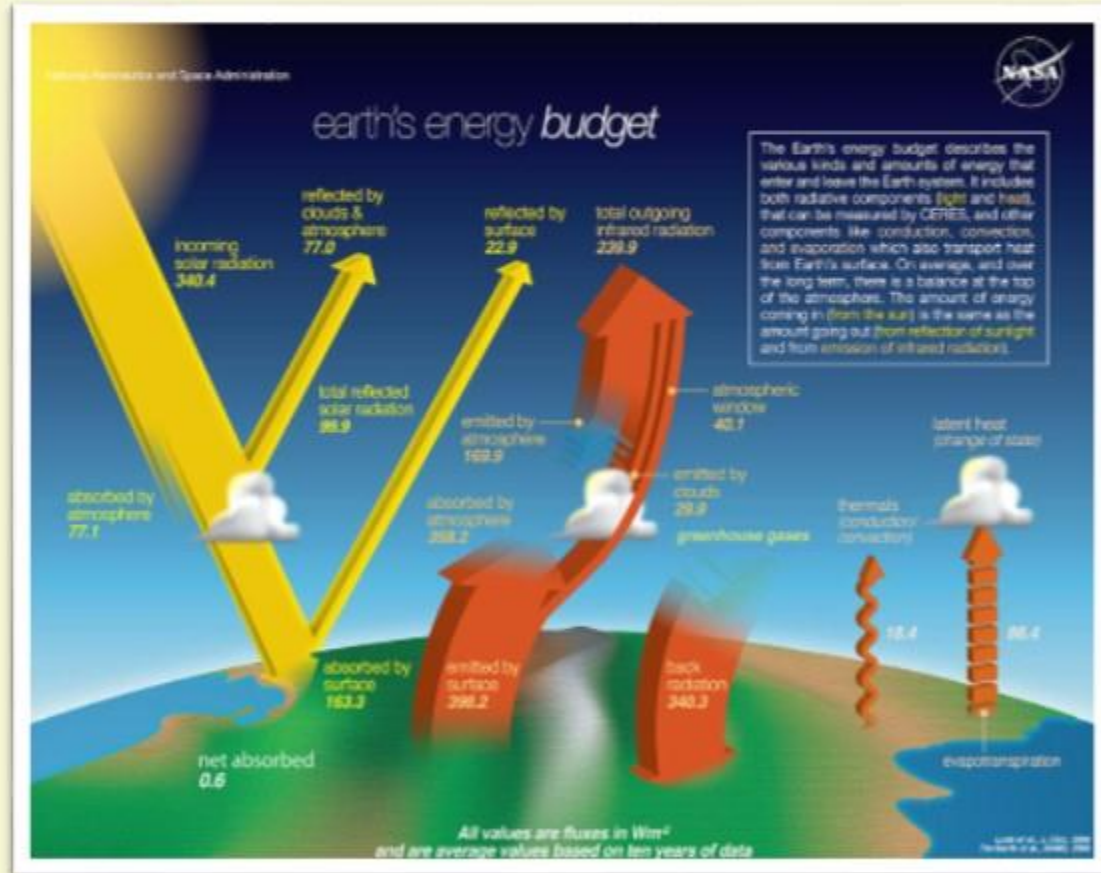
ADVECTION

- 1 The transfer of heat through **horizontal movement of air** is called **advection**.
- 2 Horizontal movement of the air is relatively more important than the vertical movement. In **middle latitudes**, most of diurnal (day and night) variation in daily weather are caused by advection alone.
- 3 In tropical regions particularly in northern India **during summer season local winds called 'loo'** is the outcome of advection process.





HEAT BUDGET OF THE EARTH



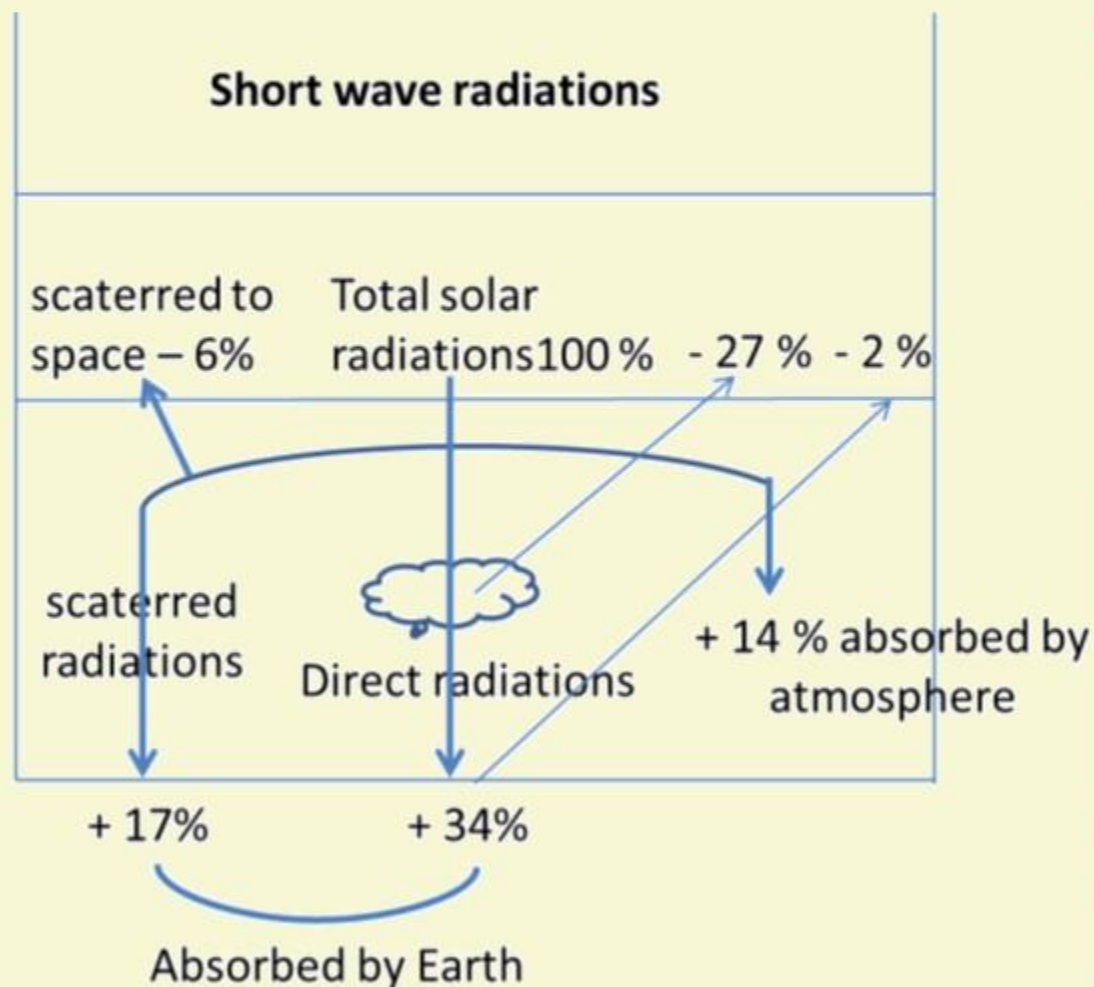
- 1 The earth as a whole **does not accumulate or lose heat**.
- 2 The amount of heat received from the sun is returned to space, thereby **maintaining constant temperature at the earth's surface and in the atmosphere**.
- 3 This can happen only if the amount of heat received in the form of **insolation** equals the amount lost by the earth through **terrestrial radiation**.
- 4 This balance of heat is known as **heat budget**.



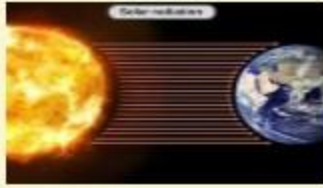
HEAT BUDGET OF THE EARTH



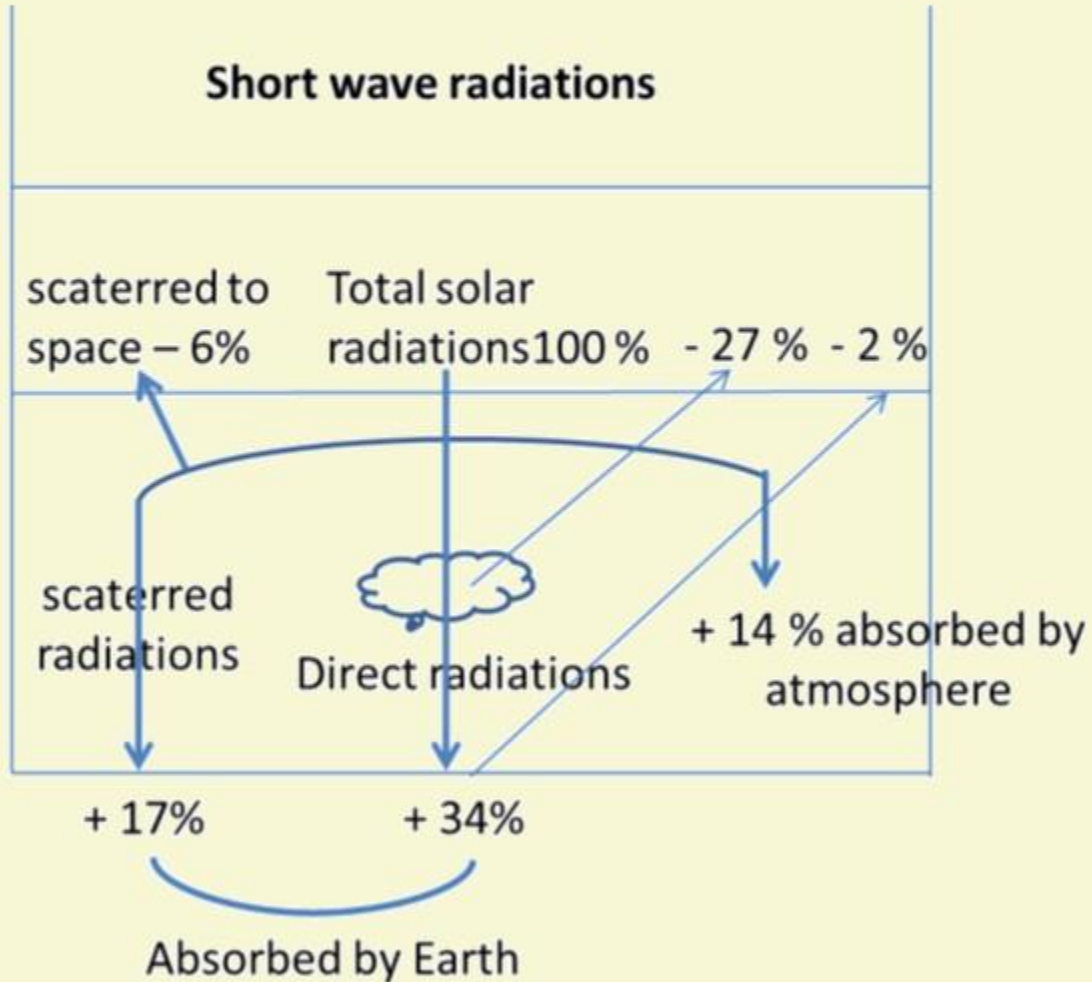
SHORT-WAVE HEAT BUDGET



- 1 Consider that the insolation received at the top of the atmosphere is 100 per cent.
- 2 While passing through the atmosphere some amount of energy is reflected, scattered and absorbed. Only the remaining part reaches the earth surface.
- 3 Roughly 35 units are reflected back to space even before reaching the earth's surface. Of these, 27 units are reflected back from the top of the clouds and 2 units from the snow and ice-covered areas of the earth.
- 4 The reflected amount of radiation is called the albedo of the earth.

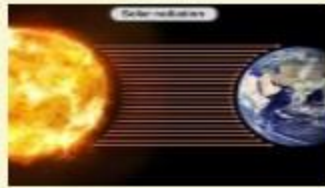


HEAT BUDGET OF THE EARTH



SHORTWAVE HEAT BUDGET

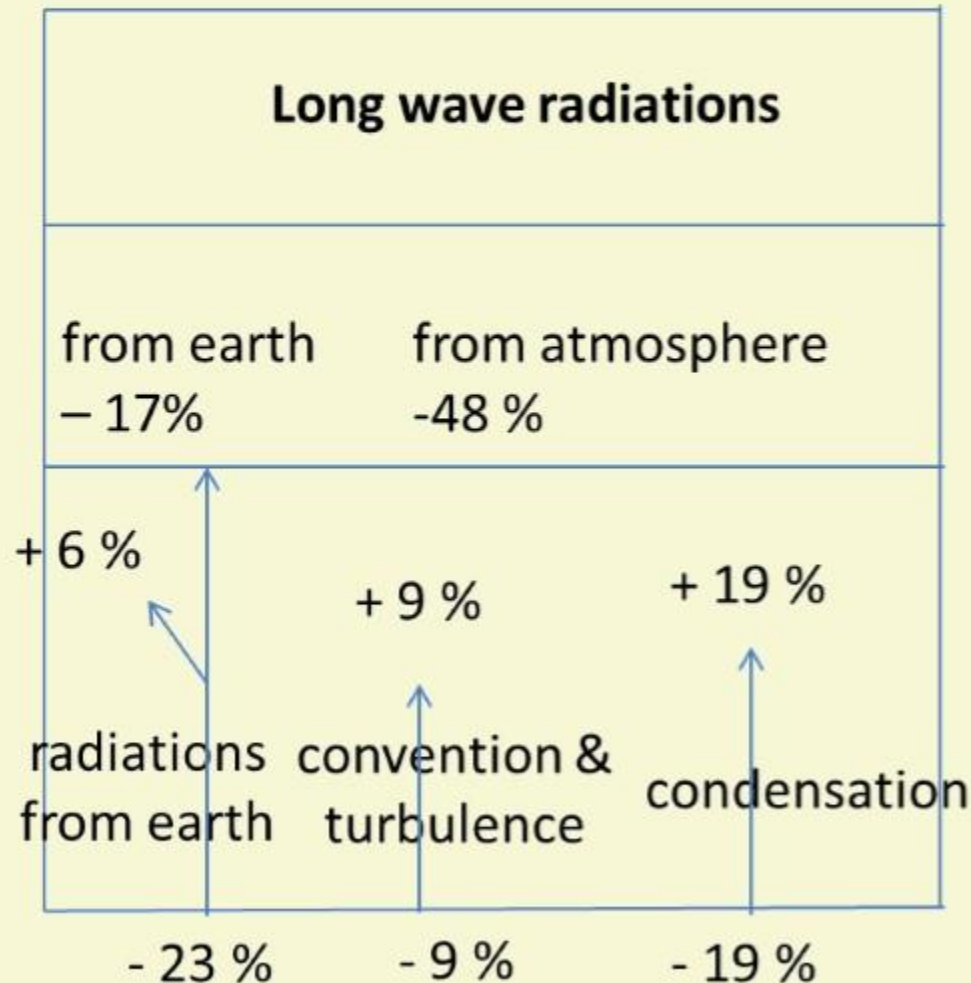
- 1** The remaining 65 units are absorbed, 14 units within the atmosphere and 51 units by the earth's surface.
- 2** The earth radiates back 51 units in the form of terrestrial radiation.



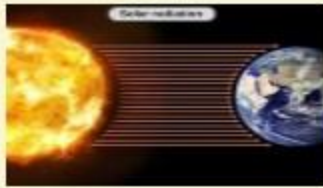
HEAT BUDGET OF THE EARTH



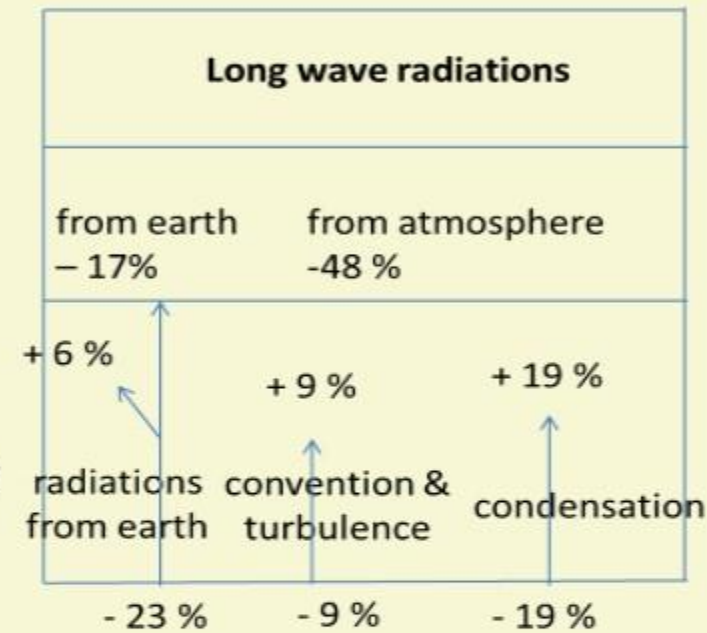
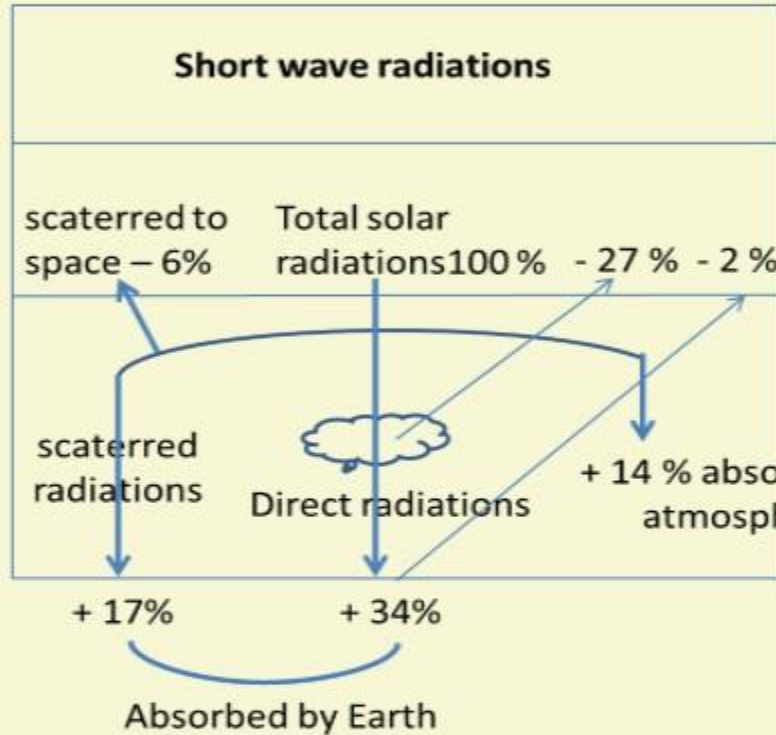
LONGWAVE HEAT BUDGET



- 1 The earth radiates back 51 units in the form of terrestrial radiation.
- 2 Of these, 17 units are radiated to space directly and the remaining 34 units are absorbed by the atmosphere (6 units absorbed directly by the atmosphere, 9 units through convection and turbulence and 19 units through latent heat of condensation)
- 3 48 units absorbed by the atmosphere (14 units from insolation + 34 units from terrestrial radiation) are also radiated back into space.

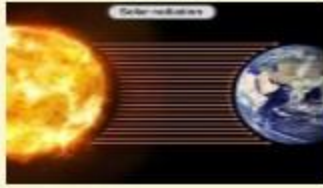


HEAT BUDGET OF THE EARTH



1 Thus, the total longwave radiation returning from the earth and the atmosphere respectively is $17+48=65$ units which balance the total of 65 units received from the sun.

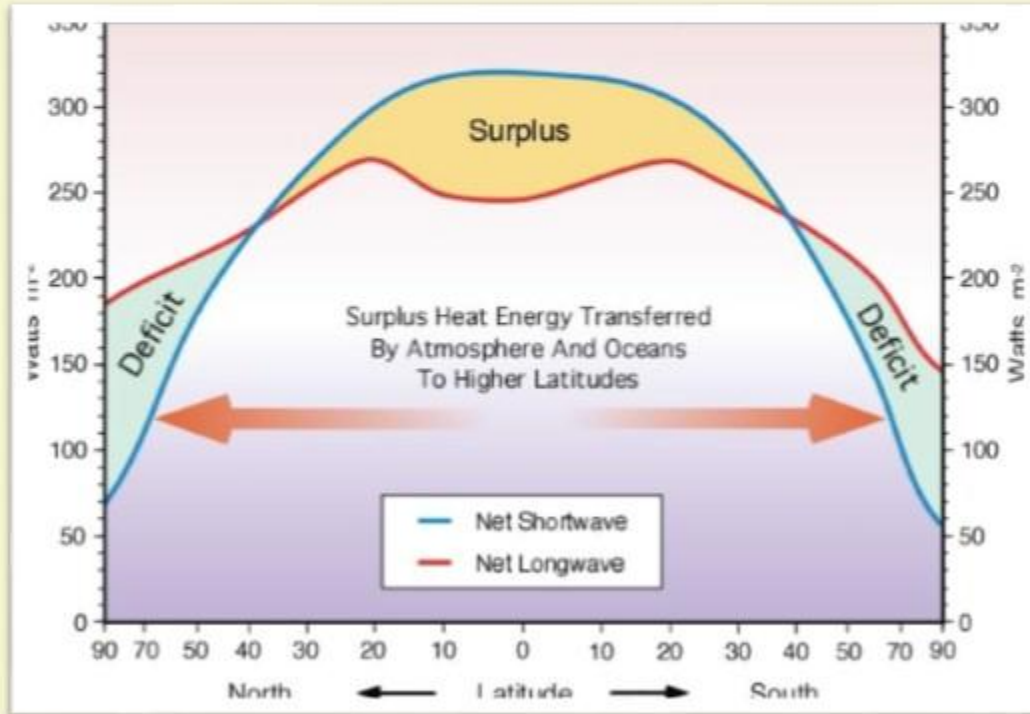
2 This is termed the heat budget or heat balance of the earth. This explains, why the earth neither warms up nor cools down despite the huge transfer of heat that takes place.



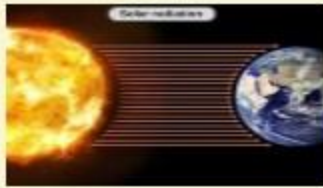
HEAT BUDGET OF THE EARTH



NET RADIATION AND LATITUDINAL HEAT BALANCE



- 1 There are variations in the amount of radiation received at the earth's surface.
- 2 The figure shows that there is a surplus of net radiation balance between 40 degrees north and south and the regions near the poles have a deficit.

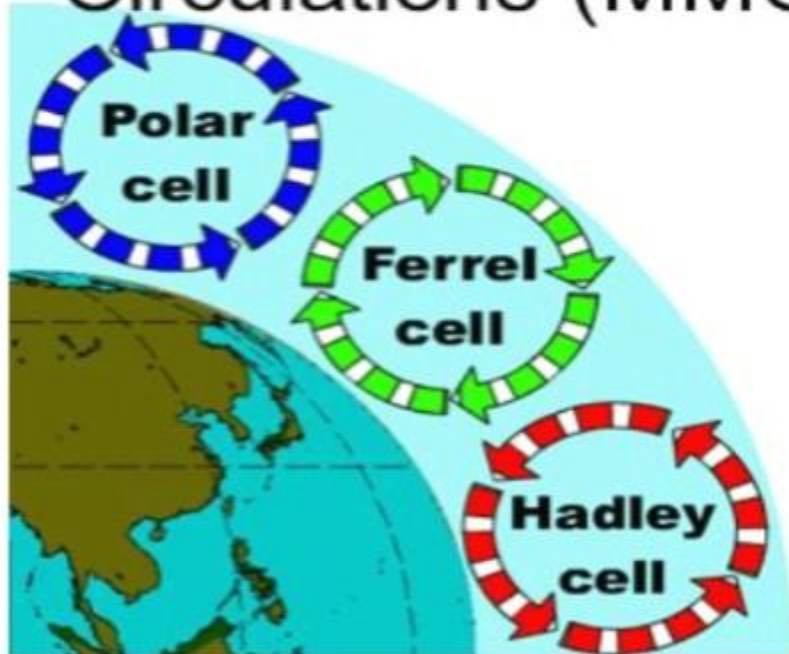


HEAT BUDGET OF THE EARTH



MERIDIONAL TRANSPORT OF HEAT

Mean Meridional Circulations (MMC)



- 1 To balance the energy rich and deficit areas, it is clear that there must be a 2-way transfer of heat: from the earth's surface to the atmosphere and from the equator to the poles.
- 2 This transport of heat from equator towards the poles via the atmosphere is known as Meridional transport of heat.
- 3 The meridional transport of heat energy in the form of sensible heat is accomplished by the atmospheric circulation and ocean currents.

POSSIBLE MAINS QUESTION

TOPIC:

GS Paper I : Salient features of world's physical geography

Explain the heat budget of the Earth. Discuss how the human-induced emissions harm this balance.(12 Marks)

HINTS

- 1 Explain heat budget.
- 2 Human emissions- GHG, PM, O₃
- 3 Forcing like CO₂, soot furthers positive feedback and thus global warming
- 4 Holocene to Anthropocene.

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